SUPPLEMENTAL MATERIAL

Table S1. The search strategies of databases.

Databases	Search strategies
MEDLINE	Search ((((("somnipathy, diabetes, cardiovascular disease, coronary heart disease, stroke, mortality "[Mesh]) OR (((((((((sleep disorder [Title/Abstract]) OR sleep disturbance [Title/Abstract]) OR sleep apnea [Title/Abstract]) OR insomnia [Title/Abstract]) OR short duration sleep [Title/Abstract]) OR somnipathy [Title/Abstract]))) AND (diabetes [Title/Abstract]) OR diabetic mellitus[Title/Abstract])) AND ((((cardiovascular disease [Title/Abstract]) OR coronary heart disease [Title/Abstract]) OR stroke [Title/Abstract]) OR (mortality [Title/Abstract]) OR death [Title/Abstract])) AND ((randomized controlled trial[Publication Type] OR non-randomized controlled trial[Publication Type] OR observational studies [Publication Type] OR randomized[Title/Abstract] OR non-randomized [Title/Abstract] OR observational [Title/Abstract]))
Web of Science	**#1** TS=(sleep disorder OR sleep disturbance OR sleep apnea OR insomnia OR short duration sleep OR somnipathy) **#2** TS=(diabetes OR diabetic mellitus) **#3** TS=(cardiovascular disease OR coronary heart disease OR stroke OR mortality OR death) **#4** TS=(randomized controlled trial OR randomized OR non-randomized controlled trial OR non-randomized OR observational) **#5** **#4** AND **#3** AND **#2** AND **#1**
Embase	**#1** 'somnipathy'/exp **#2** 'somnipathy':ab,ti OR 'sleep disorder':ab,ti OR 'sleep disturbance':ab,ti OR 'sleep apnea':ab,ti OR 'insomnia':ab,ti OR 'short duration sleep':ab,ti **#3** #1 OR #2 **#4** 'diabetes'/exp

	#5 'diabetes':ab,ti OR 'diabetic mellitus':ab,ti **#6** #4 OR #5 **#7** 'cardiovascular disease':ab,ti OR 'coronary heart disease':ab,ti OR 'stroke':ab,ti OR 'mortality':ab,ti OR 'death':ab,ti **#8** 'randomized controlled trial':ab,ti OR 'randomized':ab,ti OR 'non-randomized controlled trial':ab,ti OR 'non-randomized':ab,ti OR 'observational':ab,ti **#9** #3 AND #6 AND #7 AND #8
Cocharne library	**#1** MeSH descriptor: [somnipathy] explode all trees **#2** (somnipathy):ti,ab,kw OR (sleep disorder):ti,ab,kw OR (sleep disturbance):ti,ab,kw OR (sleep apnea):ti,ab,kw OR ('insomnia):ti,ab,kw OR (short duration sleep):ti,ab,kw **#3** #1 OR #2 **#4** (diabetes):ti,ab,kw OR (diabetic mellitus):ti,ab,kw **#5** (cardiovascular disease):ti,ab,kw OR (coronary heart disease):ti,ab,kw OR (stroke):ti,ab,kw OR (mortality):ti,ab,kw OR (death):ti,ab,kw **#6** #3 AND #4 AND #5

Table S2. The reason of exclusion for the 7 reports according to Figure 1.

No	Study	Reason
1	Hamilton GS, Naughton MT. Impact of obstructive sleep apnoea on diabetes and cardiovascular disease. Med J Aust. 2013 Oct 21;199(8):S27-30.	No relevant outcomes
2	Resnick HE, Redline S, Shahar E, et al. Diabetes and sleep disturbances: findings from the Sleep Heart Health Study. Diabetes Care. 2003 Mar; 26(3):702-709.	No relevant outcomes
3	Stadler S, Jalili S, Schreib A, et al. Association of sleep-disordered breathing with severe chronic vascular disease in patients with type 2 diabetes. Sleep Med. 2018 Aug; 48:53-60.	No relevant outcomes
4	Rissling MB, Gray KE, Ulmer CS, et al. Sleep Disturbance, Diabetes, and Cardiovascular Disease in Postmenopausal Veteran Women. Gerontologist. 2016 Feb;56 Suppl 1(Suppl 1):S54-66.	No relevant outcomes
5	Zhang Y, Lin Y, Zhang J, et al. Association between insomnia and type 2 diabetes mellitus in Han Chinese individuals in Shandong Province, China. Sleep Breath. 2019 Mar;23(1):349-354.	No relevant outcomes
6	Williams CJ, Hu FB, Patel SR, et al. Sleep duration and snoring in relation to biomarkers of cardiovascular disease risk among women with type 2 diabetes. Diabetes Care. 2007 May;30(5):1233-1240.	Follow-up<12 months or unknown
7	Li L, Gong S, Xu C, et al. Sleep duration and smoking are associated with coronary heart disease among US adults with type 2 diabetes: Gender differences. Diabetes Res Clin Pract. 2017 Feb;124:93-101.	Follow-up<12 months or unknown